EXP NO: 10

NMAP live Host Discovery using ARP Scan, ICMP Scan and TCP/UDP ping Scan in TryHackMe

Aim:

To discover live hosts on a network using Nmap host-discovery Scans — ARP scan, ICMP Scan, and TCP/UDP ping scan — and to understand which protocol operates at which layer of the TCP/IP model and when each one applies.

procedure:

1. Log in to TryHackMe and open the Nmap Live Host Discovery room.
2. Complete Task 1–3 covering the introduction, subnetworks, and TCP/IP Layer Concepts.
3. Deploy the lab network in Task 4 and start the Attack Box. Also note target IP address
4. Execute target — listing
5. Execute an ARP scan and record the number of live hosts detected.
6. Execute ICMP ECO, Time stamp and Address — Mask scans
7. Execute TCP SYN, TCP ACK and UDP ping Scans and compare the results.
8. Record the hosts detected and observe which Scan techniques are filtered.
9. Execute reverse — DNS commands
10. Complete Tasks 5–9 in the Room.

Commands:

A) ARP Host Discovery

nmap -PR -sn TARGET

Discovers live hosts on the local subnet using ARP requests

B) ICMP Echo Scan

nmap -PE -sn TARGET

Output:

ARP Scan.

Nmap Scan report for 10.200.6.1
Host is up
Nmap done: 256 IP addresses (1 hosts up)
Scanned

ICMP Echo scan:

Nmap scan report for 10.200.6.1
Host is up
:
Nmap done: 256 IP addresses (1 hosts up)
Scanned

ICMP Timestamp Scan:

Nmap Scan report for 10.200.6.1
Host is up
:
Nmap done: 256 IP addresses (2 hosts up)
Scanned.

ICMP Address-Mask Scan

Nmap done: 256 IP addresses (0 hosts up)
Scanned

TCP SYN Ping

Nmap Scan report for 10.200.6.250
Host is up
:
Nmap done: 256 IP addresses (2 hosts up)
Scanned

TCP ACK Ping

Nmap scan report for 10.200.6.250
Host is up
:
Nmap done: 256 IP addresses (2 hosts up)
Scanned

Uses ICMP Echo Requests to identify reachable hosts

2. ICMP Timestamp Scan

nmap -PP -sn TARGET

Uses ICMP Timestamp to host discovery

3. ICMP Address Mask Scan

nmap -PM -sn TARGET

C) TCP and UDP Host Discovery

1. TCP and SYN Ping

nmap -PS -sn TARGET

Sends TCP SYN to identify hosts

2. TCP ACK Ping

nmap -PA -sn TARGET

Sends TCP ACK probes to identify reachable hosts.

3. UDP Ping

nmap -PU -sn TARGET

Uses UDP probes to identify reachable hosts

Result:

Thus, live hosts were investigated using Nmap ARP, ICMP, TCP, and UDP host discovery techniques.

UDP ping

Nmap scan report for 10.200.6.250
Host is up
Nmap done: 256 IP addresses (2 hosts up) Scanned.

